From GeoAI to environmental impact assessment: A systematic review of uncertainty continuity and digital twin architecture for the circular built environment

Introduction · Related work · Methods · Results · Discussion

Complete-text working draft, without abstract. Numerical coding was reproduced; all-search eligibility and complete semantic recoding are not closed. Empirical frequencies and conceptual architecture diagrams are explicitly distinguished.

1. Introduction

The circular built environment depends on understanding how existing buildings become future material resources and how alternative recovery pathways affect environmental outcomes. Material production is a substantial source of greenhouse gas emissions, while material efficiency and changes in construction materials can alter future resource demand and carbon balances (Hertwich, 2021; Pauliuk et al., 2021; Churkina et al., 2020). Circularity therefore requires more than increasing recycling rates: it requires connecting knowledge of existing stocks with technically feasible recovery and environmentally meaningful substitution. Reviews of circular construction identify diverse practices, conceptual frameworks and implementation barriers across building life cycles (Joensuu et al., 2020; Çimen, 2021; Mhatre et al., 2021; Akhimien et al., 2021). These perspectives motivate a connected assessment of the evidence supporting circular decisions, including the uncertainty that accompanies each transformation from observation to environmental consequence.

Integrated assessments provide the starting point for examining this challenge. Mastrucci and colleagues connect GIS-based building-stock characterisation and a spatiotemporal database with life-cycle assessment of end-of-life scenarios to support urban waste-management decisions (Mastrucci et al., 2017). Ajayebi and colleagues integrate building geometries and construction histories with product-stock estimates and embodied-carbon mapping for prospective urban mining (Ajayebi et al., 2020). Their temporal characterisation and reuse-oriented inventory do not, however, establish a complete forecast of recoverable outflows. Pei and colleagues review integrated building-material-stock and LCA workflows encompassing data collection, stock modelling and result evaluation (Pei et al., 2024). These studies establish connected pathways while differing in temporal treatment, recovery modelling and environmental scope.

The Forest to Building Digital Framework additionally connects geospatial, production and building information with actor-based LCA and decision support in a wood-construction supply chain (Morganti et al., 2025). Its boundary differs from urban mining, but it illustrates connections across technical and organisational subsystems. These integrated approaches motivate examination of complete workflows before decomposing their transformations into analytical interfaces. The resulting stages must be justified through comparison, rather than imposed as a universal five-stage sequence. The research boundary extends from spatial information to material stocks and flows, recovery, environmental assessment and decision support.

GeoAI expands observational capabilities through the interpretation of heterogeneous spatial data (Wang et al., 2024; Mai et al., 2025). Mapping built-structure mass and combining geospatial and image analysis for demolition-waste modelling illustrate links between observation and material accounting (Frantz et al., 2023; Bogoviku and Waldmann, 2021). These developments complement established GIS methods; they do not imply that earlier integrated frameworks implemented GeoAI. Downstream estimates depend on inferred attributes, their resolution and validity, and assumptions translating them into material quantities.

Within-archetype variability and correlations between building size and material intensity complicate this translation (Dai et al., 2025; Miatto et al., 2023). Integrated BIM--GIS approaches organise stock information across representations (Elshaboury et al., 2025). Yet more detailed inventories do not automatically establish

how uncertainty in inferred attributes affects estimated mass. Interfaces therefore concern both numerical propagation and consistency of the entities, classifications and assumptions linking models.

Subsequent transformations introduce temporal and scenario dependencies. Lifespan assumptions, renovation and future development affect when stocks become flows and their composition (Miatto et al., 2017; Göswein et al., 2021; Tang et al., 2021; Arehart et al., 2022). Dynamic LCA further highlights changing assessment conditions through time (Slavkovic and Stephan, 2025). The analytical question is whether such transformations preserve upstream uncertainty and dependencies or substitute independently specified scenarios.

Recovery assessment evaluates whether released materials or components can enter subsequent use pathways. Material passports can support end-of-life information management, while probabilistic reusability assessment explicitly addresses technical eligibility (Honic et al., 2021; Rakhshan et al., 2021). Accordingly, an estimated outflow is not equivalent to a reusable supply. Condition, contamination, separation and recovery assumptions may change both the eligible quantity and its assigned destination. Environmental assessment then translates these pathways into impacts. Assessment methods for component reuse, life-cycle databases and design-stage assessment practices influence how burdens and benefits are represented (De Wolf et al., 2020; De Wolf et al., 2023; Roberts et al., 2020). Uncertainty must also be distinguished from variability when interpreting environmental results (Michiels and Geeraerd, 2020). The recovery--impact interface therefore connects uncertain recovery outcomes with assumptions about substitution, system boundaries and environmental characterisation.

Computational connection alone does not establish continuity of uncertainty. Exchanging an output need not transfer its uncertainty, dependencies or conditions of validity. Distinguishing uncertainty from variability is already important within LCA (Michiels and Geeraerd, 2020); across models, these distinctions must survive changes in entities, spatial aggregation, temporal horizons and methods. Here, uncertainty continuity denotes the traceable retention or explicit revision of uncertainty information and relevant dependencies across linked transformations. Existing integration and dynamic-assessment reviews provide foundations (Pei et al., 2024; Slavkovic and Stephan, 2025). The question pursued here is how far interface-level continuity is demonstrated, where it breaks down and what structural requirements follow, without presuming that uncertainty has been universally neglected.

Digital twins offer a potential architectural basis for investigating these connections. Semantic construction twins and broader characterisations of digital twins emphasise relationships between physical entities, digital representations and updating processes (Boje et al., 2020; Jones et al., 2020). Construction reviews document enabling technologies and applications, while also revealing variation in implementation and maturity (Opoku et al., 2021; Tuhaise et al., 2023). Wider reviews and built-environment definition studies likewise caution against treating the digital-twin label as evidence of a uniform capability (Sharma et al., 2022; Abdelrahman et al., 2025). Building-twin frameworks provide further ways to organise acquisition, integration, modelling and analysis (Zhang et al., 2025). For circular material assessment, the relevant issue is therefore which demonstrated capabilities can preserve meaningful relationships between observations, material states, future flows and environmental calculations.

Uncertainty-aware digital-twin research provides complementary methodological resources. Reviews of uncertainty quantification and software risk, and probabilistic graphical formulations of predictive twins, show how uncertainty can be represented alongside linked models and updated information (Lin et al., 2021; Kapteyn et al., 2021). Their application to circular building flows, however, requires examining domain compatibility. This review investigates persistent entity identification, information provenance, probabilistic representations, dependency management, updating and decision communication as candidate architectural functions. Spatial scale, temporal validity and stakeholder information requirements are treated as cross-cutting dimensions. Neither continuous sensing nor increased model detail is assumed to reduce uncertainty automatically; the proposed architecture must instead make supported transformations and unresolved limitations inspectable.

The objective is to derive an evidence-traceable digital-twin architecture for uncertainty continuity from spatial and GeoAI-supported information to environmental impact assessment and decision support. Integrated workflows are first compared to define analytical interfaces from their documented transformations. The review then connects evidence on uncertainty at these interfaces, domain-specific digital-twin applications and transferable uncertainty-related mechanisms. Four questions guide the synthesis:

RQ1. How do existing integrated frameworks connect spatial information, material stocks and flows, recovery, environmental assessment and decision support, and which interfaces emerge from their comparison?

RQ2. What uncertainty information and dependencies are transferred, transformed, updated or lost across these interfaces, and how is this demonstrated?

RQ3. Which demonstrated digital-twin capabilities address the identified interface requirements, and under what spatial, temporal and information-governance conditions?

RQ4. How can these findings be synthesised into an architecture that distinguishes directly supported components, cross-study synthesis and proposals addressing documented gaps?

Section 2 reviews the conceptual foundations and related work. Section 3 specifies the connected search, selection, interface extraction and synthesis methods. Section 4 presents the evidence analysis and architectural synthesis. Section 5 discusses interpretation, limitations and validation priorities. The contribution is a justified architectural synthesis; operational effectiveness remains a question for subsequent validation.

2. Related work and conceptual foundations

Circular material assessment connects information produced for different purposes: observations describe buildings, stock models estimate material content, dynamic models predict releases, and recovery and environmental assessments evaluate subsequent use. Integrated studies establish this connection. Mastrucci et al. (2017) link geospatial stock characterization with life-cycle assessment of end-of-life scenarios; Ajayebi et al. (2020) connect building information with spatial and temporal product-stock estimates; and Pei et al. (2024) review stock–LCA integration. Their boundaries differ, but together they motivate comparison of the transformations linking observations to circular decisions.

This comparison organizes the review around five activities: observing and inferring building characteristics (S1), estimating material stocks (S2), modelling stock change and outflows (S3), evaluating recovery and subsequent use (S4), and assessing environmental consequences and supporting decisions (S5). These labels identify comparable activities, not a universal architecture implemented by every study. Workflows may combine, omit or revisit them. GeoAI extends the observational activity through spatial inference, while image-based demolition-waste modelling illustrates its connection to material accounting (Wang et al., 2024; Bogoviku and Waldmann, 2021).

The interfaces introduce substantive assumptions. S1→S2 associates building attributes with material composition or intensity. Schiller et al. (2019) show that transferring composition indicators between German and Japanese residential buildings requires harmonization and attention to building type and context; coefficients are not universally interchangeable. Building-size–intensity relationships and lifespan assumptions also influence stock and release estimates (Miatto et al., 2023; Miatto et al., 2017). At S2→S3, a stock becomes a dated flow. Shared attributes and parameters can therefore create dependencies between upstream and downstream estimates that should not be discarded during model coupling.

At S3→S4, an estimated outflow must be distinguished from an available secondary resource. Material passports address information needed for end-of-life management, while probabilistic reusability assessment evaluates technical suitability (Honic et al., 2021; Rakhshan et al., 2021). Condition, separation and quality

influence the recoverable quantity and its destination. S4→S5 then connects those outcomes to environmental accounting, including the treatment of subsequent use and substitution (De Wolf et al., 2020). These boundaries remain analytically relevant even when one model performs several activities.

Uncertainty and variability have different meanings in this chain (Michiels and Geeraerd, 2020). Incomplete knowledge about an attribute, variation among buildings and alternative future scenarios cannot automatically be interpreted as equivalent probability distributions. We use *uncertainty continuity* to mean the traceable retention or explicit revision of uncertainty information, dependencies and validity conditions across transformations. Continuity does not require a distribution to remain unchanged: propagation, new observations or aggregation may legitimately alter it. The question is whether that alteration is documented and usable downstream. Missing reporting alone does not establish that uncertainty was lost in the implementation.

Digital-twin research supplies a complementary perspective on connected representations and updating. Systematic reviews characterize DT concepts and construction applications, while semantic construction twins emphasize information integration (Jones et al., 2020; Boje et al., 2020; Opoku et al., 2021; Tuhaise et al., 2023). These functions require separate assessment: a geometric model need not maintain a data connection, and an updated representation need not propagate uncertainty. Abdelrahman et al. (2025) distinguish real-time operational twins from longer-term decision-support applications. For material assessment, a survey, renovation or recovery inspection may be the relevant update trigger; a continuous sensor stream is not itself evidence of a valid connected assessment.

Uncertainty-aware DT research offers more specific mechanisms. Kapteyn et al. (2021) demonstrate probabilistic modelling, calibration and updating in an aircraft-related application, while Lin et al. (2021) review uncertainty quantification and software risks. Their mechanisms motivate examination of state representation, dependencies and revision procedures, but do not demonstrate continuity throughout the circular-material chain. Transfer requires matching entities, quantities, time horizons and validation conditions. Likewise, a scalable software architecture does not establish that material assumptions remain valid across buildings, districts and regions.

The review connects these literatures through their interfaces: material-flow research identifies what is transformed; uncertainty research identifies what information must be retained or revised; DT research identifies capabilities for linking evidence and calculations. Spatial support, temporal validity and stakeholder information requirements qualify these connections. Building on stock–LCA, uncertainty and DT reviews, the analysis distinguishes directly documented connections, cross-study synthesis and proposals addressing gaps. Its contribution is an evidence-traceable architectural synthesis, rather than a newly implemented or validated end-to-end twin.

3. Materials and methods

3.1. Review design: a shared interface as the unit of synthesis

The review connects evidence on material-assessment workflows, uncertainty and digital-twin capabilities through a shared analytical unit: a documented interface between activities. Search families provide complementary routes to this evidence, rather than separate reviews with unrelated conclusions. The design addresses four linked questions: which transformations connect S1–S5; what happens to uncertainty at those transformations; which DT capabilities are relevant under the reported conditions; and how the findings support an evidence-traceable architecture. PRISMA 2020 guides reporting of study selection and PRISMA-S guides documentation of searches (Page et al., 2021; Rethlefsen et al., 2021). These reporting guidelines do not replace the extraction and synthesis procedures specified below.

For publication p and interface i→j, the extraction record is

E(p,i,j) = {entity, input, transformation, output, uncertainty-in, uncertainty-out, dependencies, DT-function, context, evidence}

. Entity identifies the building, component or material represented; context includes spatial support, temporal validity and information-user requirements; evidence contains the publication identifier and page, section, equation or figure locator. DT-function is recorded as not applicable or not reported where appropriate. Thus a non-DT material-flow study and a DT mechanism study can be compared through an explicit information requirement without claiming that they implement the same system. Figure 1 shows this connected workflow; Supplementary Figure S1 specifies the analytical joins.

3.2. Information sources and search families

The database searches cover Scopus, Web of Science Core Collection and IEEE Xplore. Base, also denoted S, refers to stage-specific building/material-flow searches. U denotes uncertainty and DT denotes digital twin. The domain families are Base, Base+U, Base+DT and Base+DT+U. General DT and DT+U searches provide definition and transferable-mechanism evidence, with their non-domain origin retained. Supplementary searches for integrated workflows are not restricted by publication year or publisher. Exact filters are documented per search execution rather than assumed identical across historical runs.

Scopus searches use TITLE-ABS-KEY; Web of Science searches use Topic, which also includes Keywords Plus. IEEE field selections and split-query variants are retained in the execution records. The logical search structure is Q(i,k) = B(i,k) AND S(i) AND optional(U) AND optional(DT), where k identifies the query version. The version index matters because some historical DT searches use a broader domain block than earlier Base searches. The logical diagram is therefore not evidence that every historical query forms a strict subset of one identical Base query.

Supplementary Table S1 lists the concepts used to organize retrieval. The complete archived strings, including exclusions and database adaptations, remain in Supplement S1 and the recovered S1–S3 query files. Missing historical execution details are flagged in the audit rather than reconstructed as executed searches. New explanatory examples in this section are not presented as additional database searches.

The search-block table, counting equations and reproducibility checks are provided in the supplement. Continuity distributions count assessed paper–boundary contexts, with one status per context. Stage–function edges count distinct publications carrying the exact paired code; memberships overlap and are not summed as unique papers.

For clarity, the four domain combinations are B AND S(i); B AND S(i) AND U; B AND S(i) AND DT; and B AND S(i) AND DT AND U. General DT+U omits the building-stage restriction. Broad U terms improve retrieval but do not establish uncertainty analysis: sensitivity, calibration or robustness must be interpreted from the methods. Membership is recorded at publication level after deduplication. Actual overlaps are calculated from identifiers, not inferred from query labels or summed database counts (Rethlefsen et al., 2021).

3.3. Eligibility, deduplication and selection

Each publication is screened against its intended evidential role. Domain studies must describe a relevant activity or transformation in building-material assessment. U evidence must identify an affected quantity, assumption or decision and explain its treatment; a generic uncertainty statement is insufficient for a mechanism claim. DT evidence must describe representations, data connections, updating or analytical functions. Architecture papers can provide conceptual evidence without being relabelled operational implementations. External-domain mechanisms and secondary reviews are retained separately as complementary evidence. A paper need not address all five stages or demonstrate successful uncertainty continuity to be relevant.

Original database exports are preserved. DOI values are normalized; records without DOI are checked using native identifiers and bibliographic information. Similar titles, conflicting DOIs and related conference/journal versions require adjudication. Duplicate publications are distinguished from multiple reports of one study. Multiple search memberships are retained without multiplying the unique-publication denominator.

Selection proceeds through identification, deduplication, title/abstract screening, full-text retrieval, full-text eligibility and inclusion. Full text is checked for identity, completeness and methodological detail. Abstract-only material and incomplete previews do not support full-text mechanism extraction. Reports that cannot be obtained or adequately assessed do not enter the final synthesis. Their records remain in the selection audit: non-retrieval is distinguished from exclusion after assessment, and available but unread reports remain pending rather than being assigned a scientific exclusion reason. No target number of included publications is imposed.

Full-text exclusions have a primary reason, with secondary notes where needed: outside the material-flow domain; no relevant DT function for a DT-specific synthesis; no substantive U treatment for a U-specific synthesis; or insufficient methodological evidence for the intended analytical role. A report can be ineligible for the strict triple intersection while remaining relevant to a complementary stream. Reasons and evidence locators are preserved. PRISMA reporting follows these distinctions (Page et al., 2021).

3.4. Interface extraction and uncertainty coding

Stage assignment is based on the operation actually performed. An edge S(i)→S(j) requires a documented use of an upstream output in a downstream activity; mentioning two stages does not establish an edge. Non-adjacent links and feedback are retained. Table 2 defines the common questions applied across the four adjacent interfaces. These are extraction prompts, not findings that every study implements the listed transformations.

Interface	Transformation to inspect	Uncertainty information to trace	Candidate DT function to investigate
S1→S2	Observations/attributes to material quantities	Classification uncertainty, geometry, material intensities and their dependencies	Identity, semantic mapping, provenance and data-quality representation
S2→S3	Inventory to time-dependent release	Inventory uncertainty, lifetimes, survival and scenario dependencies	State/time representation, linked dynamic models and versioning
S3→S4	Released material to recoverable supply	Timing, condition, separation yield, quality and route eligibility	Condition updates, component tracking and recovery-service linkage
S4→S5	Recovery routes to impacts/decision	Yield, substitution, allocation, impact factors and joint assumptions	Assessment-service coupling, scenario provenance and decision communication

The extraction distinguishes uncertainty source, nature, representation and operation. Parameter, model and scenario uncertainty are recorded separately from spatial/temporal variability where the study permits this distinction (Michiels and Geeraerd, 2020). Representations include distributions, samples, intervals, scenarios and qualitative quality statements. Dependencies include reported covariance, correlation, shared parameters and explicit independence assumptions. Missing dependence information is not coded as independence.

The interface state uses non-exclusive operation codes: transferred, transformed, updated, newly introduced, or demonstrably discarded. A separate reporting status identifies insufficient description. Reduced uncertainty requires a stated comparison and metric; improved prediction accuracy alone does not establish reduction. Source statements are distinguished from reviewer interpretation. Each mechanism claim requires a locator and a short evidence note. Scale, temporal resolution, update trigger, validity period and stakeholder information roles are extracted as conditions of the interface, not as disconnected descriptive categories.

3.5. Linking interface requirements to DT capabilities

DT coding records the authors' definition, physical referent, digital representation, data connection, update direction/frequency, analytical service and demonstrated maturity (Jones et al., 2020; Boje et al., 2020; Abdelrahman et al., 2025). Real-time operation and bidirectional control are not imposed as universal requirements. Nor is a BIM/GIS model automatically treated as an operational DT. The distinction between probabilistic modelling and demonstrated asset–twin updating is informed by uncertainty-aware DT research (Kapteyn et al., 2021; Lin et al., 2021).

For each interface, an evidence chain links: observed limitation → information requirement → reported DT capability → compatibility conditions → supported architectural connection. Compatibility is checked against the entity, variable meaning, unit, spatial support, temporal horizon and dependency representation. For example, a general-domain Bayesian updating mechanism is not assigned to S3–S4 merely because both involve uncertainty; the proposed connection must explain what recovery-related state would be updated, by which observation and under which assumptions. If that connection is inferred across studies, it is explicitly coded as synthesis rather than a demonstrated domain implementation.

3.6. Relational analysis and architectural synthesis

The same interface records generate three connected matrices: interface × uncertainty operation; interface × DT capability; and DT capability × uncertainty operation. Every populated cell resolves to supporting publication and evidence identifiers. A fourth join table links all three dimensions and records whether the connection occurs within one study or is synthesized across studies. No complete S1–S5 chain is inferred simply by concatenating unrelated papers. Supplementary Figure S1 shows the shared record that prevents the three matrices from becoming independent analyses.

For a given eligible corpus C, the cell count is n(i,u,d) = the number of distinct publications in C supporting interface i, uncertainty operation u and DT capability d together. Direct within-publication counts and cross-study synthesis are reported separately; inferred links do not inflate empirical frequencies. Coverage of a coded property is its distinct supporting-publication count divided by the number of eligible, assessed publications for that analysis. The denominator and unreported cases accompany each figure. Multilabel coding means marginal totals can exceed the unique-publication total. This treatment follows the need to distinguish study counts from category participation, illustrated in Falsafi et al. (2025) (Falsafi et al., 2025).

Descriptive outputs include evidence heatmaps, overlap plots and interface networks. A line represents a coded connection, not causality; thickness represents a specified count, not effectiveness. Temporal counts summed across databases are labelled database occurrences unless record-level deduplication supports unique-publication counts. Statistical tests are not assumed valid for overlapping categories, and visual differences are not labelled significant without an appropriate test design. Heterogeneous outcomes are not pooled in an effectiveness meta-analysis.

Architectural elements are classified as directly supported, synthesized from compatible studies, or proposed in response to a documented gap. The architecture thus links each module and connection to its evidential basis and to the interface requirement it addresses. A reported limitation, missing reporting and an absence in the retrieved corpus remain distinct. The resulting structure is a review-based conceptual synthesis; operational effectiveness and cross-scale generalization require subsequent validation.

3.6.1. Exact positive-code configurations (D03)

For the non-review DT–U subset, each publication is represented by z(p)=(z1,...,z7), where zk=1 for a reported R-coded mechanism and 0 otherwise. The seven dimensions are probabilistic representation, Bayesian state/parameter inference, explicit dependency transport, residual/shift diagnostics, online adaptation, interval-coverage assessment and uncertainty-linked alerts/actions. The original C and NR states are preserved separately: zero in this projection does not establish absence. Publications assigned to the review/reference-architecture family are omitted from this configuration analysis.

For each observed binary pattern v, its exact-intersection frequency is h(v)=sum_p 1[z(p)=v]. Each publication contributes to exactly one pattern, and sum_v h(v)=N. Marginal mechanism frequency is m(k)=sum_v v(k)h(v). Columns are sorted by decreasing h, then number of positive codes, with a deterministic binary-pattern tie-break. This is an UpSet-style exact-set display, not clustering, a maturity ranking or an inferred causal sequence. Identical positive-code patterns do not establish identical implementations; implementations may differ in context and in their C/NR states.

3.6.2. Source–method associations within the same transition context (D04)

Source and method labels are joined on the compound key (publication_id, context), never on publication alone. Let A(q) be the set of counted pairs carrying source q and A(m) those carrying method m. The cell count is

$$n(q,m)=|A(q) \text{ intersection } A(m)|$$

and the row denominator is

$$N(q)=|A(q)|$$

. Colour encodes

$$P(m|q)=100\ n(q,m)/N(q)$$

, while bubble area is 150 n(q,m) points squared. Empty source rows have an undefined percentage, not zero. Rows and columns are ordered by their assessment frequencies, with codebook order resolving ties.

Because labels are multilabel, percentages within a row need not sum to 100%. The join establishes co-occurrence within one assessed context, not that the method was specifically applied to that source. For example, a context containing two sources and two methods yields four co-occurrence cells; a source-specific treatment claim requires an additional explicit evidence relation. Source labels are not automatically mapped to epistemic or aleatory uncertainty. Prediction-error diagnostics and scenario comparison are not automatically probabilistic propagation.

3.6.3. Joining the four outputs into transition-based architectural evidence

D01 establishes the recorded continuity state of a boundary; D04 identifies accompanying sources and methods; D02 locates reported domain functions; D03 provides general-DT mechanism configurations. The architectural join uses the boundary record as its anchor and stores: boundary_id, material variable and unit, uncertainty source/representation, continuity state, method, DT function, update trigger, spatial/temporal validity, evidence locator and support class. Within-publication links are distinguished from compatible cross-study synthesis and proposed additions. General DT configurations are not relabelled S1–S5 implementations.

For each boundary, the results will report the assessed publication count, context count, distribution of continuity states, documented variable transformations, candidate DT functions and the evidential status of each proposed connection. A transition diagram can therefore show what is preserved, transformed, newly introduced or insufficiently reported without interpreting a stage–function edge as uncertainty transport. Missing B34 coverage remains visible. A conceptual missing link is not filled by reallocating O5 assessments to B45 or by treating a review as an independent experiment.

The architectural synthesis additionally maps each configuration family to its proposed uncertainty responsibility, material-interface application and required output. The extraction separates a source-supported concept from our allocation of that concept to S1–S5. Unresolved descriptors retain that status. A mapped capability is therefore a design requirement unless a source demonstrates the corresponding operation; candidate connections in the configuration diagram are not counted as empirical observations.

3.7. Evidence appraisal, traceability and reporting status

Appraisal considers data provenance, explicit assumptions, dependency treatment, validation, scenario conditions and reproducibility. Synthetic demonstrations, empirical tests and conceptual proposals are distinguished. A universal numerical quality score is not imposed on heterogeneous designs. The decision ledger retains source, role, eligibility reason and evidence locator, while code and count inputs accompany figures. The review does not claim an independent two-human-reviewer process where none is documented.

Figure 2 is an audit-status PRISMA-style diagram for the broad Base+DT+U search, not a completed all-scenario selection flow. Its counts are read from the archived decision ledgers. Figure 3 reports adjudication within the previously selected focused sets; these are subsets with overlapping membership, not additional independent searches. The broader Base, Base+U, Base+DT and general-DT workflows cannot yet be labelled complete through Included. Pending reports are excluded from current substantive synthesis but remain visible as pending in the audit. Completion of those decisions is required before replacing these status diagrams with final submission-ready PRISMA flows. This status statement is editorial and must be updated when the review is closed, rather than removed while the work remains incomplete.

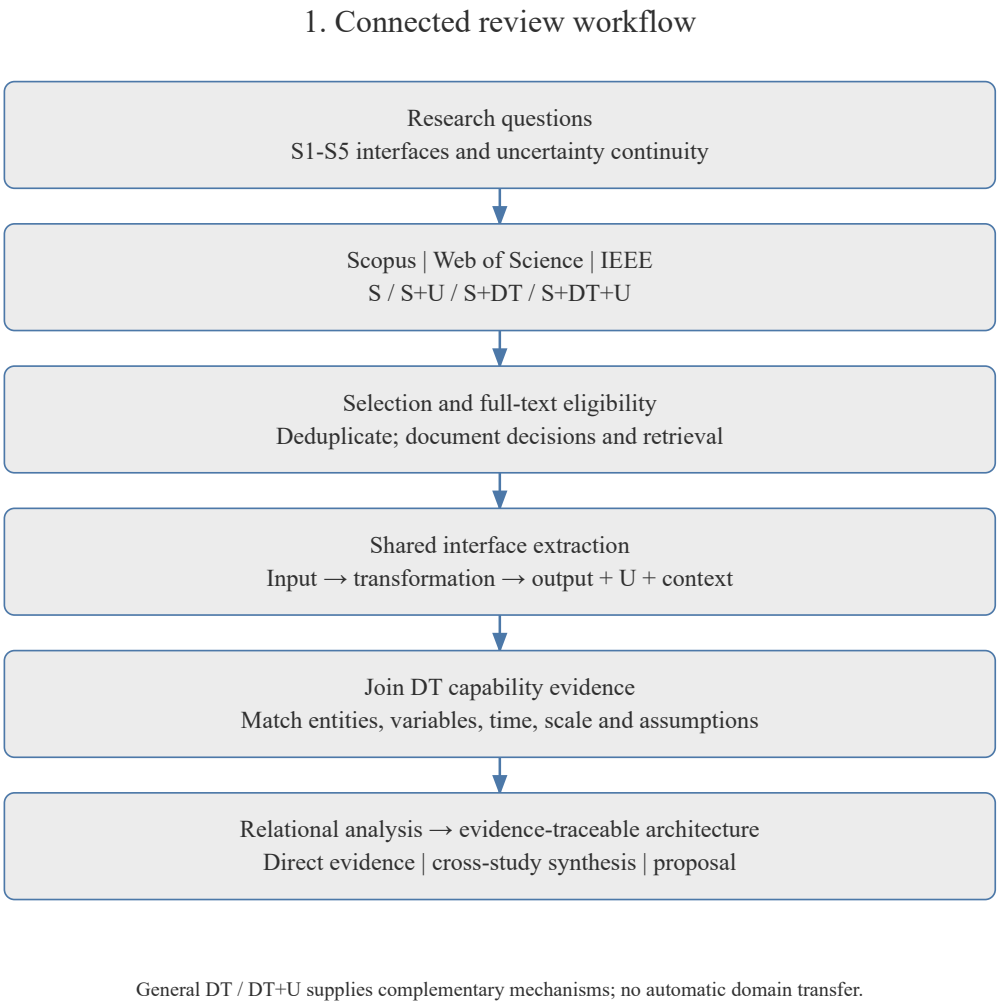


Figure 1. Connected review workflow. Conceptual methodological design.

See Supplementary Figure S1.

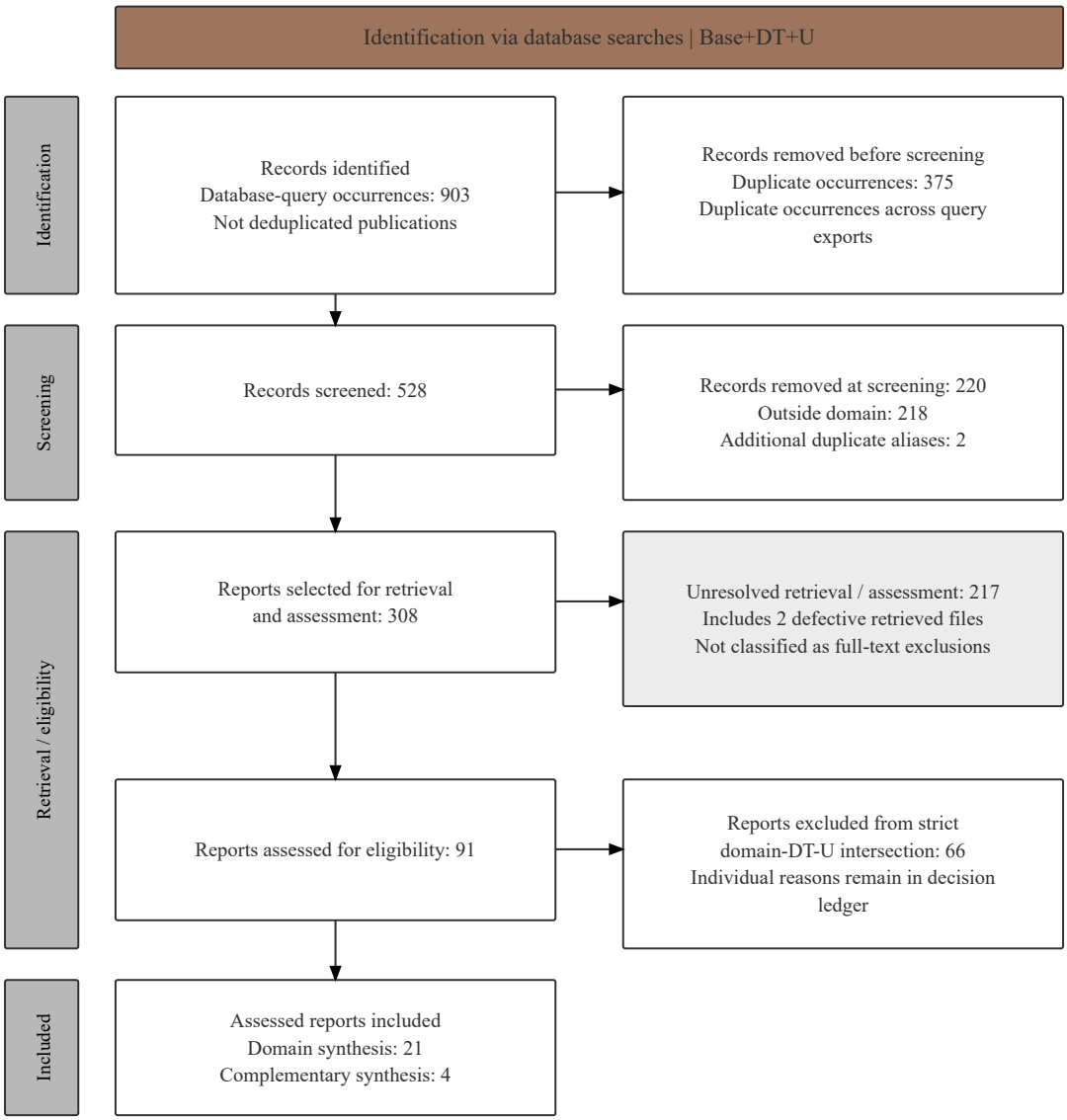


Figure 2. Broad Base+DT+U selection audit. Pending eligibility is not relabelled as exclusion.

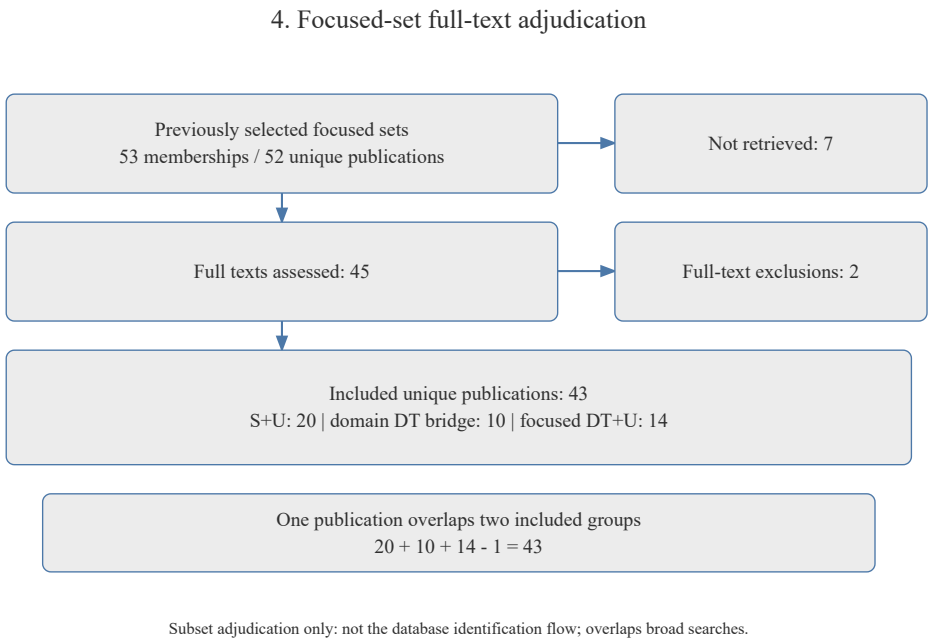


Figure 3. Focused-set adjudication; overlapping subsets, not an additional independent database flow.

4. Results

4.1. Evidence sets and the meaning of the reported percentages

The available analyses reveal an uneven connection between building information, material accounting and uncertainty-aware decisions. This finding emerges from several linked analytical subsets, rather than one interchangeable denominator. The archived S1–S3 analysis contains 17 publications with full-text or supporting-information-based coding. The current S+U relational analysis contains 20 publications contributing 23 assessed contexts. The domain DT function analysis contains nine publications with S1–S5 function codes; three adjacent-domain records are outside that graph. The general DT+U configuration analysis contains ten non-review publications. These sets support different questions and overlap in membership; their sizes must not be added to produce a review total.

All source and method percentages below were recalculated from article-level codes. Every one of the 17 legacy publications maps to an included publication in the current S+U eligibility ledger. This identity check establishes membership consistency, but does not independently validate every older interpretive code. The broader search remains incompletely adjudicated, as shown in the Methods selection diagrams. Accordingly, the reported distributions describe the named analytical subsets, not the prevalence of methods throughout the entire literature.

The percentages are frequencies of recorded evidence. For example, 64.7% means that eleven of seventeen publications carry a particular source code. It does not mean that this source contributes 64.7% of output variance. Similarly, the proportion of papers reporting Monte Carlo is not a measure of model accuracy, uncertainty magnitude or the effectiveness of uncertainty reduction.

4.2. Sources, representations and downstream outcomes

Material intensity/composition is the most frequent source in the legacy subset: 11/17 publications (64.7%). Scenario/future context follows at 7/17 (41.2%), temporal/event uncertainty at 6/17 (35.3%), and model-form/structural uncertainty and LCA inventory/factor uncertainty each at 5/17 (29.4%). Classification/prediction and hidden/missing building information each occur in 3/17 (17.6%). Observation/measurement and transferability/domain shift each occur in 2/17 (11.8%). Recovery/technical

feasibility and boundary/allocation/accounting each occur in 1/17 (5.9%). No positive spatial/logistics or demand/market-matching code is recorded in this subset. Those zeros delimit the extraction rather than establish that the wider research field ignores these problems.

The pattern places material representation and future assumptions at the centre of the observed uncertainty discussion. It also shows why a single label such as “data uncertainty” is insufficient: uncertain building attributes, material coefficients, lifetimes and environmental factors enter different transformations. Source categories do not automatically determine whether uncertainty is epistemic, aleatory or a mixture; that distinction depends on the model, observational support and decision context (Michiels and Geeraerd, 2020; Vilaysouk et al., 2021; Miatto et al., 2017; Cao et al., 2017).

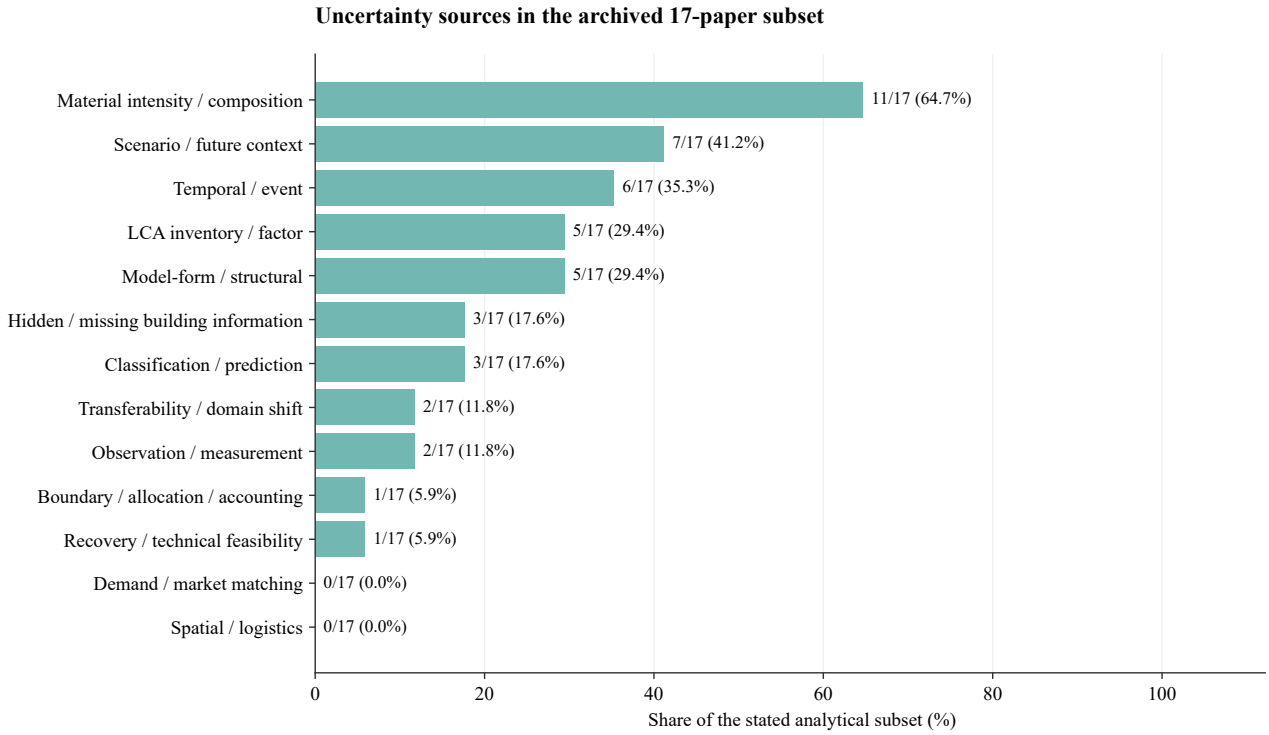


Figure 4. Recalculated source frequencies, archived 17-paper coding. Multi-label, paper denominator.

The legacy method codes record local/one-way sensitivity in 6/17 publications (35.3%); scenario ensembles and bootstrap each in 4/17 (23.5%); ranges/intervals and Monte Carlo each in 3/17 (17.6%); and probabilistic ML/parameter modelling in 2/17 (11.8%). Three publications (17.6%) carry the legacy deterministic/no-formal-UQ category. No positive Bayesian, global-sensitivity, fuzzy/possibility or analytical-error-propagation code is recorded. These are broad inherited categories: bootstrap used for training-data augmentation is not equivalent to bootstrap used to estimate an output interval, and a probability distribution assigned to an input is not proof that it was propagated. The more specific context coding used in D04 retains these functional distinctions.

See Supplementary Figure S2.

The six legacy downstream-outcome labels are also reproducible: unassessed in 9/17 publications (52.9%); transformed in 7/17 (41.2%); preserved and masked/lost each in 3/17 (17.6%); and amplified and legitimately reduced each in 1/17 (5.9%). Labels overlap because a publication may examine several boundaries or operations. In particular, a paper can analyse downstream parameter variability while leaving uncertainty from upstream inference unassessed. The outcome percentages therefore do not sum to 100%.

The frequency labels should be interpreted at the strength of their underlying evidence. The single amplified/reduced case concerns sensitivity to lifetime assumptions and spatial aggregation (Miatto et al., 2017); it does not establish that a DT reduced real uncertainty. Likewise, “transformed” can denote a change in the representation or consequences of variability, not necessarily an estimated upstream and downstream probability density. The conceptual distribution sketches make the distinctions legible, but no curve in those sketches is fitted to the reviewed studies.

See Supplementary Figure S3.

See Supplementary Figure S4.

4.3. Where continuity is documented across the four boundaries

The current boundary coding separates evidence of an uncertain input from evidence of its downstream treatment. At S1→S2, nine contexts are assessed. Four (44.4%) do not demonstrate downstream transfer; two (22.2%) provide sensitivity/representation comparisons; one each (11.1%) provides review/discussion, prepares inputs for propagation, or documents partial transfer. None is coded as quantified output uncertainty for selected inputs in this boundary-specific classification. Material-intensity distributions and prediction-error measures can therefore improve the input description without establishing a complete uncertain stock estimate (Vilaysouk et al., 2021; Zhang et al., 2022; Yuan et al., 2023).

At S2→S3, six contexts are assessed: three (50.0%) document partial transfer, two (33.3%) quantify outputs for selected uncertain inputs, and one (16.7%) does not demonstrate transfer. The stronger evidence here is consistent with the explicit use of lifetime distributions, probabilistic parameters and dynamic material-flow models (Miatto et al., 2017; Cao et al., 2017; Heeren and Hellweg, 2018; Liu et al., 2026). However, an uncertain lifetime model can coexist with deterministic upstream quantities. The evidence does not justify interpreting the entire chain as probabilistic simply because the stock-dynamics calculation uses sampling.

No S3→S4 context is assessed in this particular harmonised coding set. A percentage is therefore undefined, rather than zero percent successful propagation. The sole S4→S5 context is coded as partial transfer: 1/1, or 100% of one observation. This is a case description, not a stable estimate of methodological prevalence (Almeida et al., 2026). Seven additional contexts concern other inputs to S5. They remain separate because environmental-factor, damage or emissions analyses do not necessarily consume material-recovery outputs. Reassigning them to S4→S5 would create continuity that the extraction has not established (Olsson et al., 2025; Moon et al., 2025; Aljawhari et al., 2024; Caruso et al., 2026).

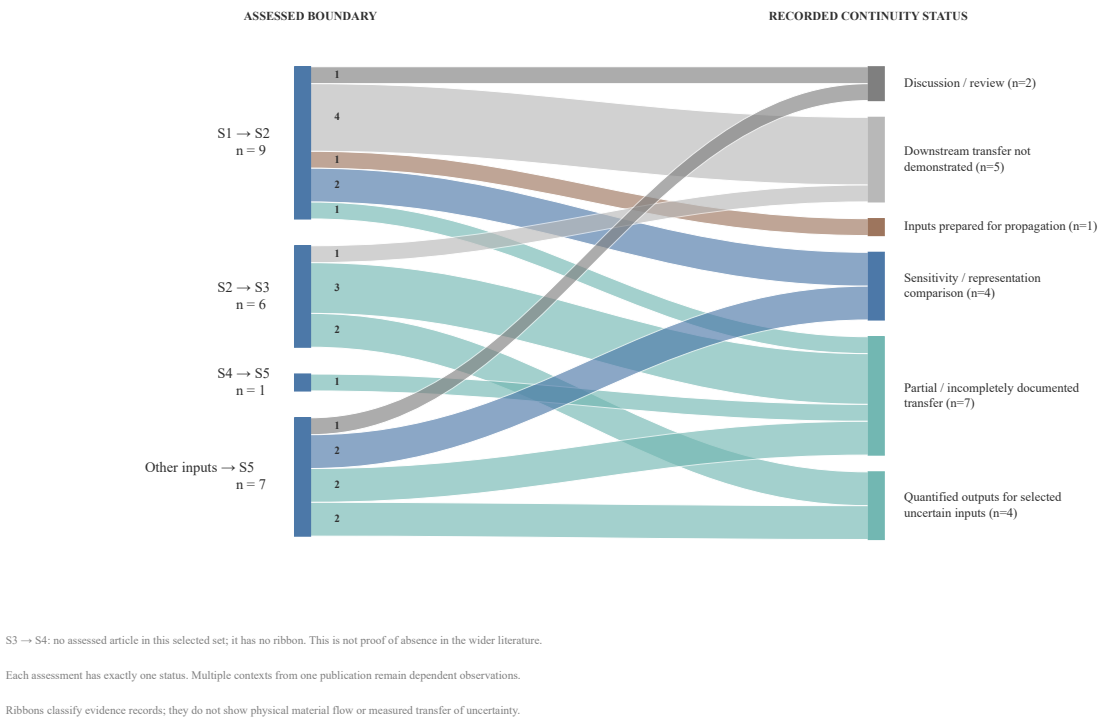


Figure 5. Current boundary–continuity coding: 23 contexts from 20 publications.

See Supplementary Figure S5.

The distribution across all 23 contexts is seven partial-transfer assessments (30.4%), five without demonstrated downstream transfer (21.7%), four sensitivity/comparison assessments and four quantified-selected-input assessments (17.4% each), two review/discussion assessments (8.7%), and one propagation-ready assessment (4.3%). These categories are mutually exclusive within a context, but the contexts are not independent studies. The classification differs deliberately from the legacy six-outcome labels: it grades what the extraction demonstrates about an interface, rather than inferring the physical fate of every uncertainty component.

4.4. Which DT functions and uncertainty mechanisms are recorded?

Within the nine domain-function records, seven publications have a positive S1 function code and seven have a positive S2 code (77.8% each). S3 and S4 each have two (22.2%), and S5 has one (11.1%). A publication may contribute to several stages. The weighted network contains sixteen positive stage–function combinations. Observation/inference, semantic material information, identity/provenance and inventory modelling dominate the recorded architecture. Presence at two stages does not establish an operational connection between them; the graph is a functional coverage map, not a measured uncertainty-transfer network.

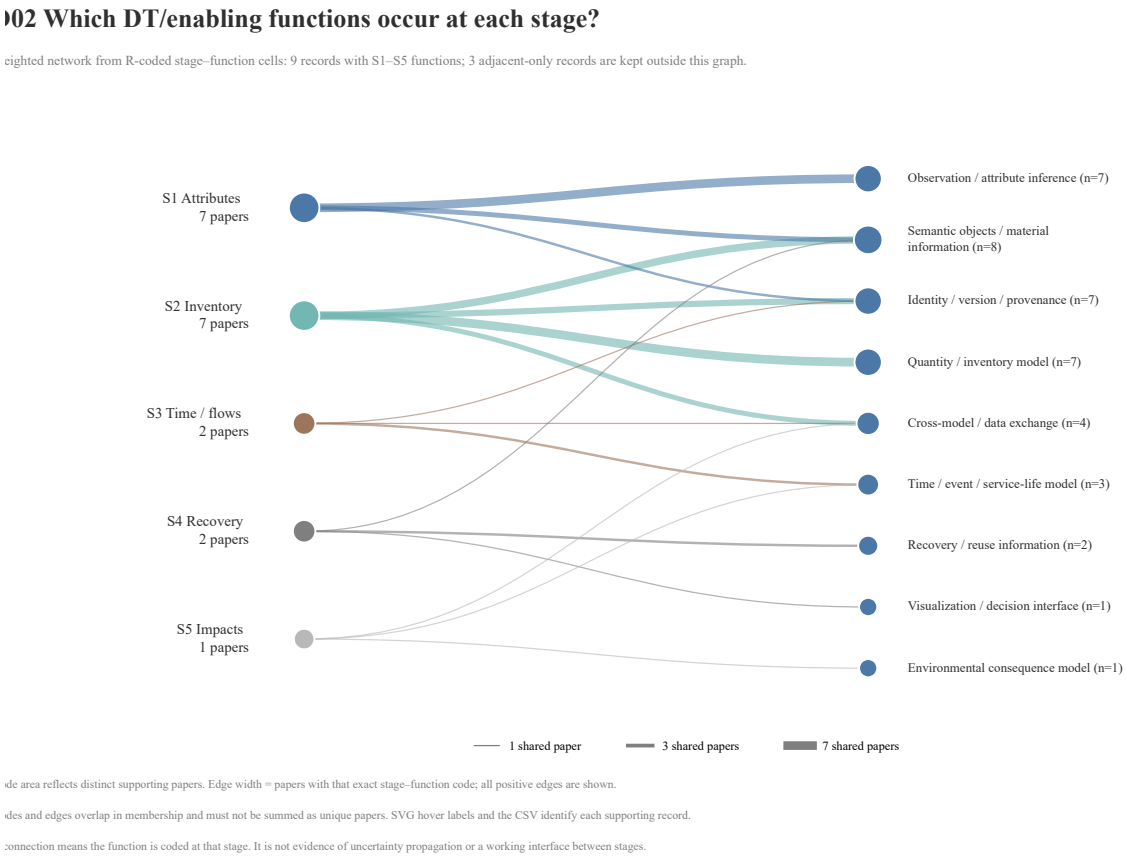


Figure 6. Stage–function co-coding; edge width is distinct supporting publications, not uncertainty flow.

The ten non-review general DT+U records show a different emphasis. Probabilistic uncertainty representation and residual/shift diagnostics each appear in eight records (80%). Bayesian state/parameter inference, online adaptation and uncertainty-linked alerts/actions each appear in seven (70%). Empirical interval-coverage assessment appears in three (30%), and explicit joint/dependence transport in two (20%). These percentages refer to recorded positive method descriptions. They neither establish successful deployment in circular construction nor prove that an unrecorded feature was absent from the implementation.

Nine exact positive-code configurations occur across the ten records. One configuration is shared by two papers; the remaining eight occur once each. This heterogeneity supports a vocabulary of composable capabilities, rather than one empirically dominant DT architecture. Especially relevant to continuity is the difference between frequently recorded prediction/update capabilities and less frequently recorded joint-dependence transport. A system can update a model and issue an alert while still losing upstream correlation information at the next model interface.

See Supplementary Figure S6.

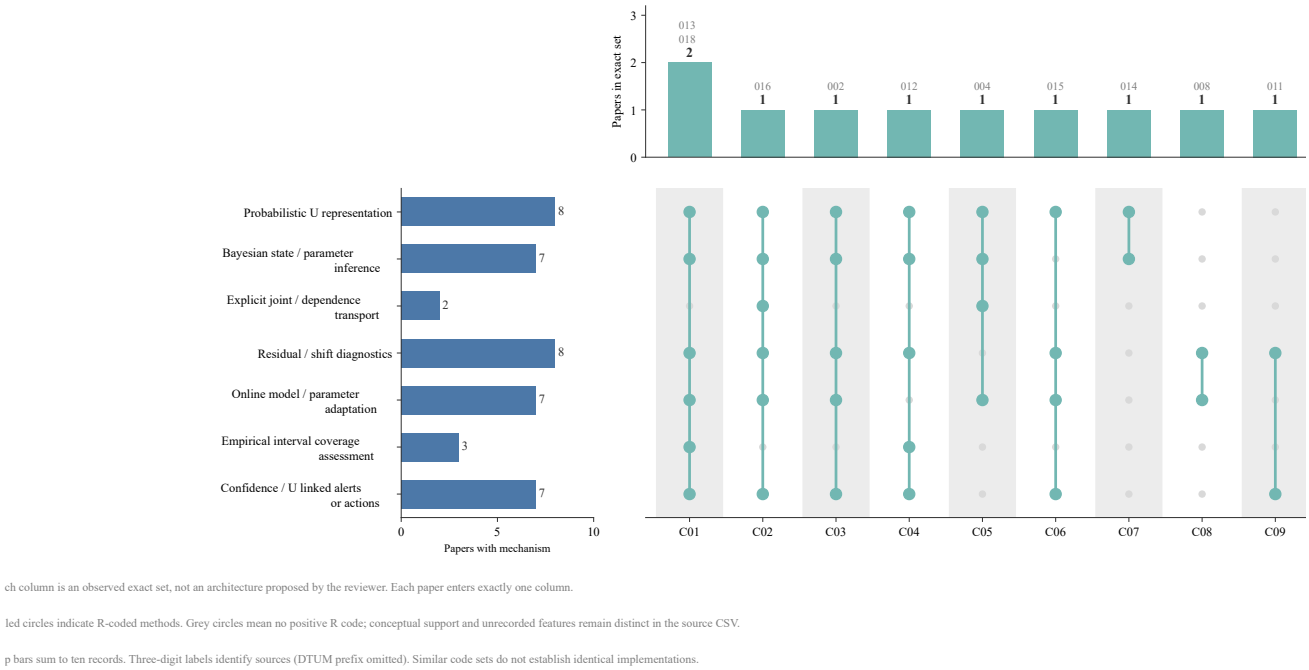
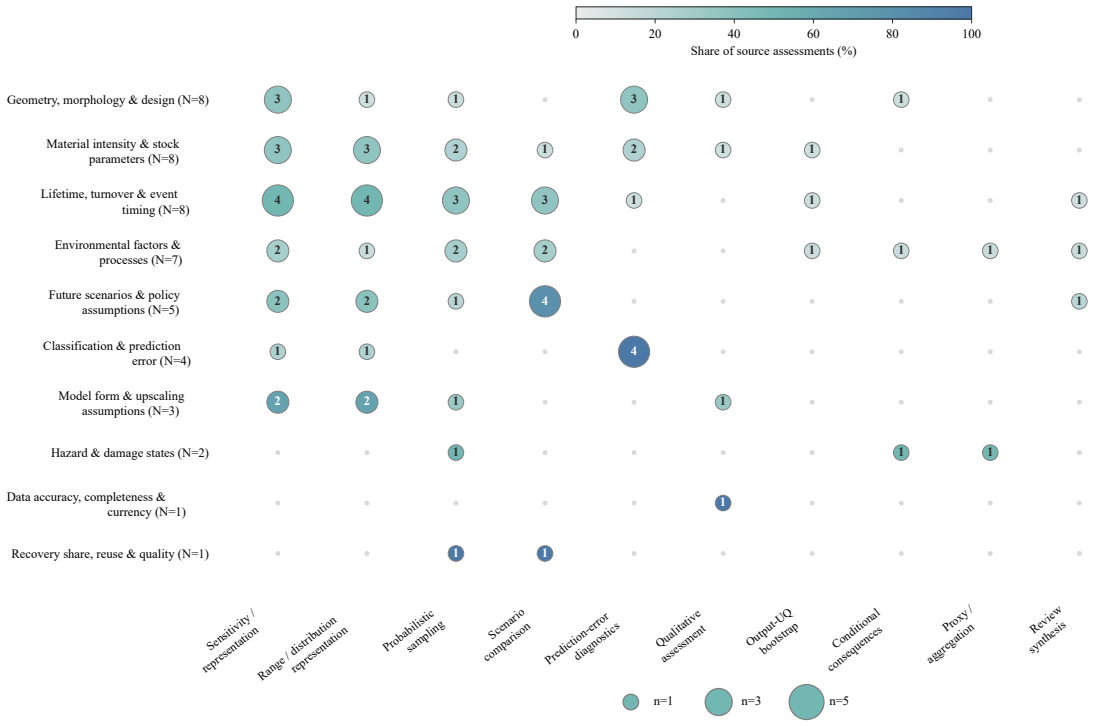


Figure 7. Exact positive-code intersections. Each publication occupies one configuration.

The source–method matrix provides a bridge between these observations. It contains 46 positive cells among 100 possible source–method cells. For geometry-related contexts, sensitivity/comparison and prediction diagnostics each occur in 3/8 contexts (37.5%). For lifetime/timing contexts, sensitivity and range/distribution representation each occur in 4/8 (50.0%). Scenario comparison accompanies future-scenario sources in 4/5 contexts (80.0%). The recovery source has only one assessed context, associated with sampling and scenario comparison. These are same-publication, same-context associations; they do not establish that every paired method was specifically applied to every paired source.

)04 Which analysis methods accompany each uncertainty source?

ɔmputed same-paper, same-context associations. Bubble area = assessment count; colour = share of that source's assessments.



ily codes within the identical publication and boundary are joined. Multiple sources/methods can occur in one assessment; cells are not additive.

e labels describe sources, not automatically epistemic/aleatory categories. A method association does not prove uncertainty transfer.

ro is no recorded association in this extraction; N is the number of assessed contexts carrying that source. All edges and supporting IDs are exported.

Figure 8. Source–method co-occurrence restricted to the same paper and context.

4.5. Locating uncertainty responsibilities within the synthesized DT structure

Figure 9 synthesizes the design, credibility-assessment and decision logic used to interpret the review findings. It separates the design brief (A), configuration choices (B), and functional allocation (C). Figures 9–11 combine reviewed DT mechanisms with an auxiliary architectural source set (R1–R16). That register includes partial-text sources and is not a separately completed systematic corpus. Sources support specific concepts; the A/B/C arrangement and its application to GeoAI–circular-material interfaces are our synthesis. The structure connects two questions: which information must remain credible for a circular decision, and which implemented component is responsible for producing, transferring or checking that information (Xiong et al., 2025; Human et al., 2023; Agrawal et al., 2023).

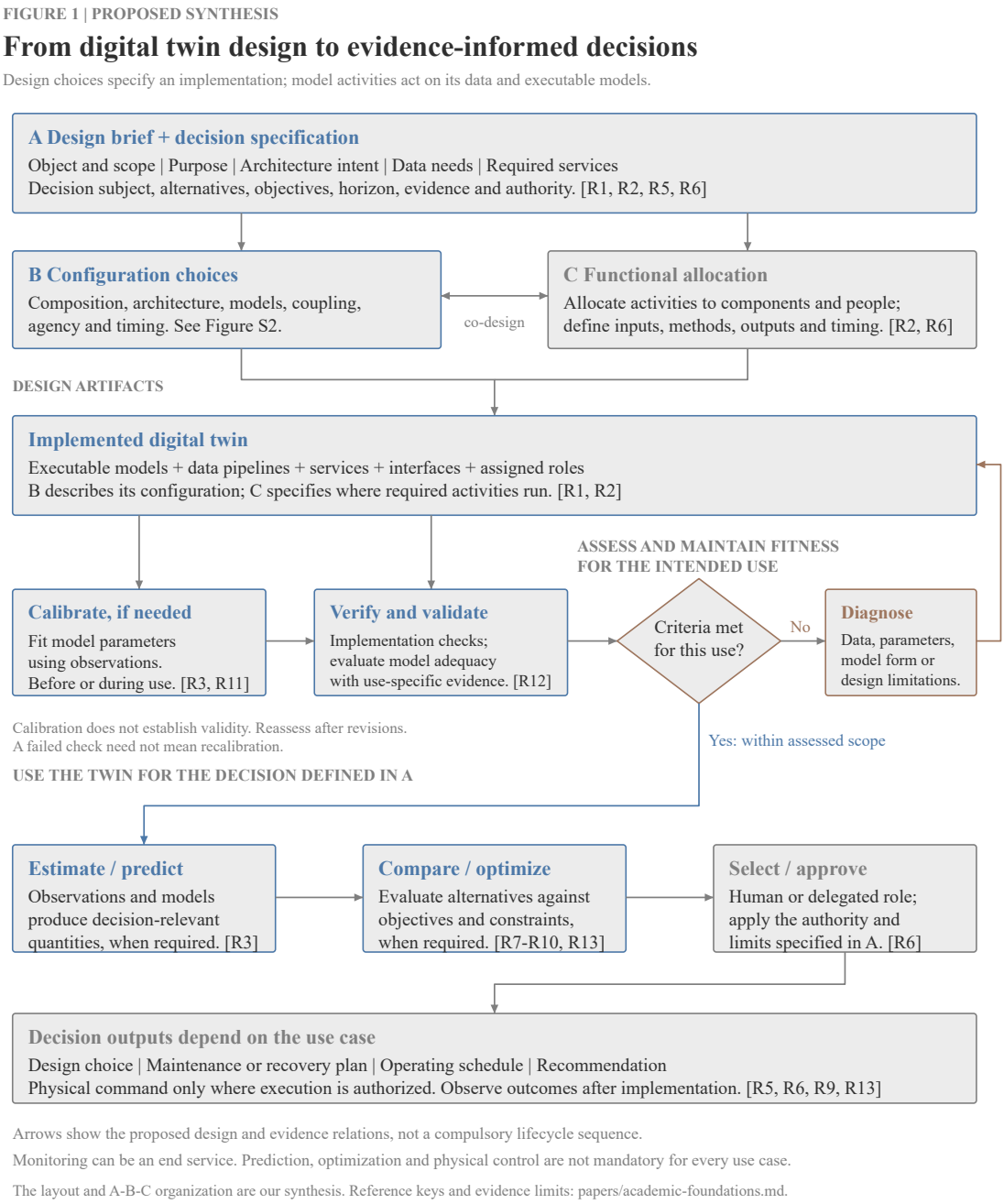


Figure 9. The synthesized design-to-use structure retained as the organizing architecture. A specifies intended use; B selects configuration; C allocates responsibilities. Calibration, validation and diagnosis operate on implemented models and data. R1–R16 are source keys, not review counts: for example, R1 = Xiong et al. (2025), R2 = Human et al. (2023), and R12 = Bitencourt et al. (2025). The supplementary register resolves every key and its reading status.

A specifies what uncertainty matters. The twinning object is a building, component or linked stock, with explicit spatial support and time horizon. The purpose determines the quantity of interest: material mass, release time, usable recovery or environmental consequence. Data requirements distinguish observed, inferred and assumed attributes, and separate fitting evidence from evaluation evidence. Service requirements identify who needs an uncertainty statement and when. Thus a district-scale stock estimate and a component-reuse decision may require different evidence even when they share source observations. These use-specific requirements follow (Shao and Helu, 2020; Agrawal et al., 2023).



Figure 10. The design brief fixes object, purpose, architectural intent, data requirements and services before configuration selection. In this review, these fields define the scope, evidence and authority under which uncertainty is assessed. The five-card arrangement is a synthesis, not a published universal taxonomy.

B determines the means of maintaining uncertainty information. Figure 11 retains all six configuration families. They are coexisting design dimensions, not successive material-flow stages. Their contribution is assessed by the operation they enable, not by assuming that a descriptor such as hybrid, federated or agent-based automatically supplies uncertainty quantification. Figure 12 expands the third card of Figure 10, the system architecture brief, into proposed services with explicit inputs, operations and checkable outputs. The six configuration families determine how these services are implemented.

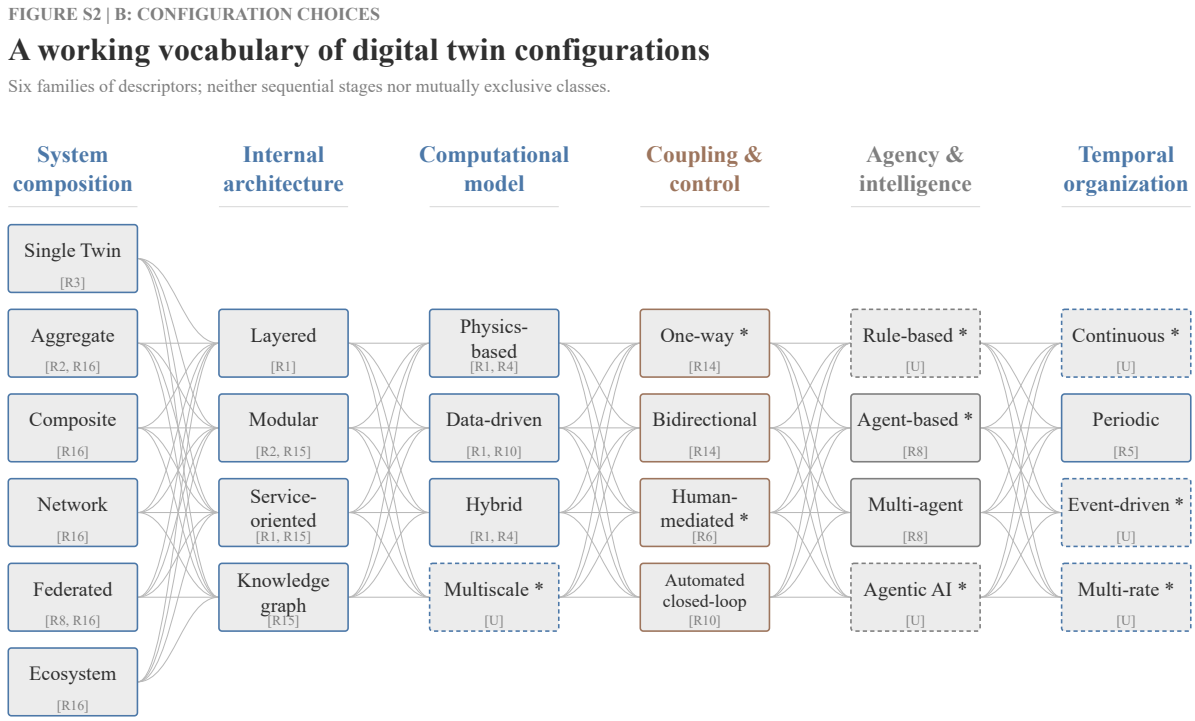


Figure 11. Configuration vocabulary used in the architectural synthesis. Connecting curves indicate candidate pairings only. Dashed entries retain their unresolved evidence status. Agent-based simulation, autonomous software agency and agentic AI must not be treated as interchangeable.

The *computational model* performs inference and propagation. GeoAI may provide geometry, attribute probabilities or prediction errors; inventory models translate these into material quantities; lifetime models produce release trajectories; recovery models apply condition and yield; environmental models evaluate consequences. An uncertainty-bearing input must be carried through these transformations rather than replaced silently by its mean. Physics-based, data-driven and hybrid models can each support this operation when their uncertain inputs, parameters, discrepancy and dependencies are specified. A probabilistic graphical formulation provides one demonstrated basis for coupled state estimation and updating, but its transfer to the full material chain remains a synthesis proposition (Kapteyn et al., 2021).

The *internal architecture* makes that operation traceable. Modules expose input/output contracts; semantic identifiers connect observations, models and material entities; provenance records the origin and version of each assumption. This is the role of integration and knowledge representation, not a substitute for probabilistic computation (Kosse et al., 2025). *System composition* determines which buildings or components are combined. For a proposed aggregation of material masses,

$$Var(\Sigma M_i) = \Sigma Var(M_i) + 2\Sigma_{i<j} Cov(M_i, M_j)$$

; preserving shared archetype or material-coefficient dependencies is therefore important.

Coupling and temporal organization determine whether estimates remain current. New survey evidence, renovation or recovery inspection can trigger a revision; provenance identifies downstream outputs requiring recalculation. Continuous updating is not necessary for every circular decision, and two-way exchange does not establish autonomous control. *Agency* determines who initiates checks, requests additional evidence, selects a model or approves an action. Agent-based simulation models interacting entities; multi-agent orchestration distributes service tasks among software actors. Agentic AI denotes goal-directed tool orchestration here, but remains unsupported as a specific configuration in this source set. Actors may request evidence or invoke checked estimators; they do not replace inference or validation. Human authority and escalation remain explicit (Kritzinger et al., 2018; Shao and Helu, 2020; Agrawal et al., 2023).

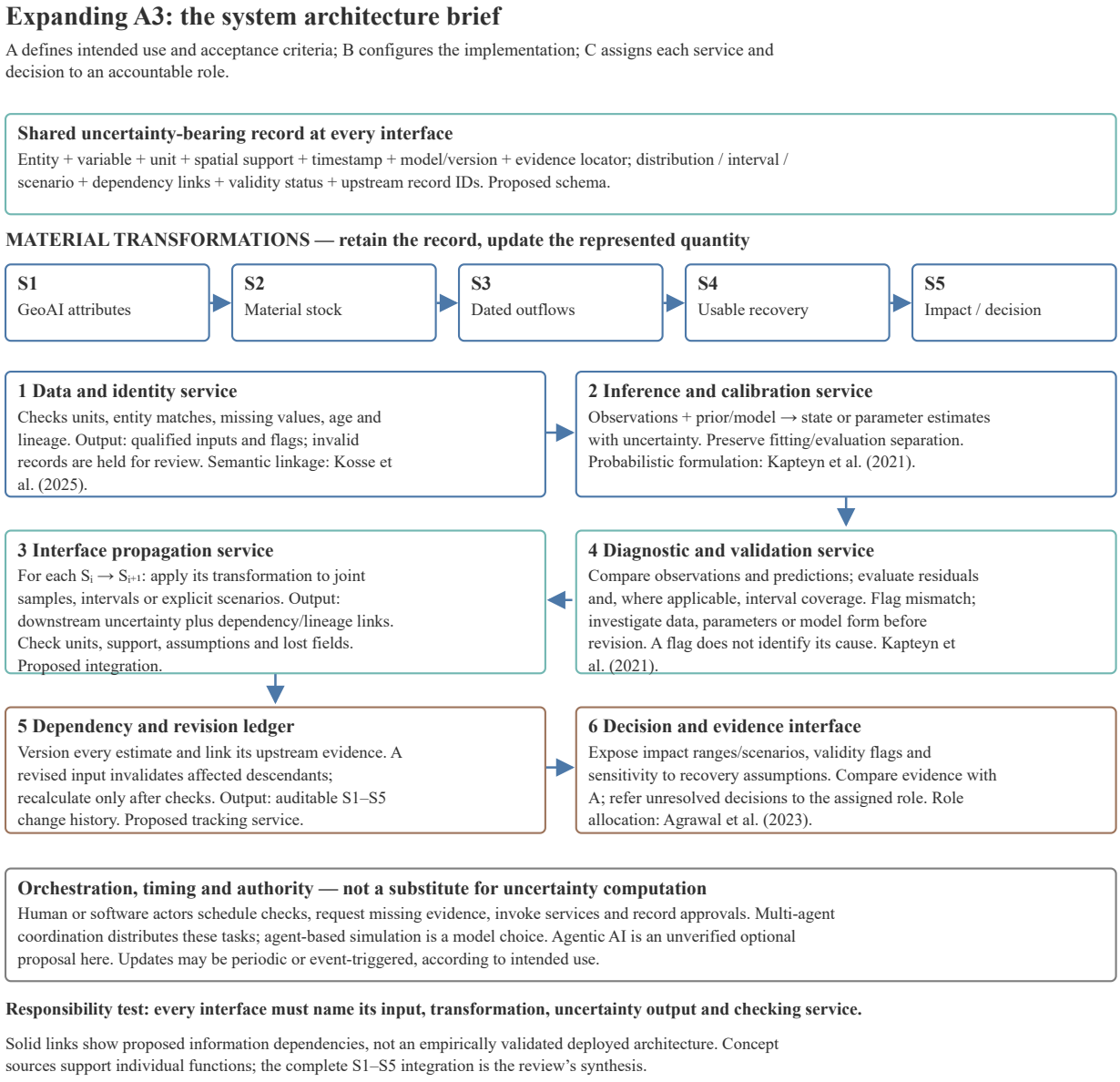


Figure 12. Technical expansion of the system architecture brief (Figure 10, third card). Numerical inference and propagation are separated from interface checks, diagnostic evaluation and orchestration. The uncertainty-bearing record crosses all four material interfaces. Service allocation and record schema are our proposed synthesis; concept-level sources do not validate the integrated implementation.

C assigns executable responsibilities and evidence gates. Detection compares an observation with its model prediction; a residual flags disagreement but does not uniquely identify its cause. Diagnosis examines data quality, parameter change, model form and operating context. Calibration updates selected parameters; validation checks performance against evidence appropriate to the intended use. Propagation carries a

distribution, interval or scenario together with its dependencies into the next model. Tracking links each revision to affected downstream outputs. Decision support communicates the resulting consequence and whether available evidence satisfies A. These functions may be distributed across components; they are not additional S stages.

Applied to the four material interfaces, this allocation asks whether GeoAI uncertainty reaches material quantities (S1→S2), whether stock and lifetime dependencies reach release forecasts (S2→S3), whether condition and separation uncertainty reach recoverable yields (S3→S4), and whether recovery and substitution assumptions reach impact estimates (S4→S5). The DT contribution is the maintained relationship between these calculations, their evidence and their revision history. The selected material-flow evidence does not yet demonstrate this complete chain. The proposed architecture specifies what would have to be implemented and assessed to close that gap.

5. Discussion

5.1. Continuity is a property of interfaces, not the presence of UQ software

The central implication is that uncertainty-aware components do not automatically form an uncertainty-continuous workflow. GeoAI can produce an error score, a stock model can use variable material coefficients, and a lifecycle model can run Monte Carlo, yet these activities may refer to different uncertain inputs. The review therefore shifts the question from whether a method appears in a paper to whether the information needed by the next transformation survives the interface. Integrated stock–flow and impact workflows motivate this question, while the transition analysis specifies how it can be examined (Bogoviku and Waldmann, 2021; Pei et al., 2024; Mastrucci et al., 2017; Ajayebi et al., 2020).

Three operations must be distinguished. Propagation computes consequences of specified uncertain inputs. Updating changes an uncertainty representation in light of observations or revised assumptions. Validation evaluates whether predictions and their uncertainty are adequate for the intended use. Running one operation does not establish the other two. In particular, a small prediction error does not establish calibrated predictive intervals, and a narrower posterior does not establish that missing mechanisms have been resolved. A digital twin is useful here because it can organise repeated observation, model revision and evidence evaluation, provided these operations are implemented explicitly (Kapteyn et al., 2021; Shao and Helu, 2020).

5.2. What must cross each material-flow boundary?

At S1→S2, the transformation links inferred building attributes to material quantities. A schematic stock calculation is

$$M = \Sigma_b A_b I_b$$

, where A denotes a relevant geometric measure and I a compatible material-intensity coefficient. The uncertainty of the product depends on both factors and their dependence. Transferring only the predicted building class or mean coefficient loses information needed by the inventory model. A useful DT interface would retain class probabilities or alternative attribute states, material-coefficient uncertainty, consistent units and the aggregation rule. It should not turn a classification confidence score into a calibrated probability without supporting evaluation.

At S2→S3, an inventory is transformed into dated outflows using age, lifetime, renovation and event assumptions. A schematic mapping

$$O(t) = \Sigma_c M_c f_c(t \mid age, scenario)$$

emphasises that the output is a flow in a time interval rather than a stock at a date. The uncertainty representation must therefore carry material quantities together with the event-time model. A new building survey can update stock information, while a renovation event may revise remaining life and outflow timing. Revisions must preserve their relationship to the previous state rather than silently overwrite the forecast (Miatto et al., 2017; Cao et al., 2017; Heeren and Hellweg, 2018; Liu et al., 2026).

At S3→S4, potential outflow becomes technically recoverable and route-specific material.

$$R = O \times \eta \times q$$

is a deliberately schematic expression: η represents yield and q a quality/acceptance factor, and their meanings must be defined for the selected reuse or recycling route. Damage, contamination and selective-disassembly evidence may change these factors and their dependence. The current absence of a coded context at this boundary makes a condition-observation and recovery-evaluation interface a justified research priority, but not a validated solution. It would be inappropriate to assign universal recovery distributions to fill that gap (Honic et al., 2021; Rakhshan et al., 2021).

At S4→S5, recovered materials acquire environmental consequences under substitution, allocation and system-boundary assumptions. A schematic impact calculation sums route-specific material quantities multiplied by environmental factors and subtracts eligible avoided burdens. The same recovered mass can lead to different results when substituted products, quality equivalence or future energy systems differ. A traceable interface must retain those conditions along with material quality and quantity. Circularity alone does not establish an environmental benefit, and a reported impact interval cannot be interpreted as end-to-end uncertainty unless upstream recovery uncertainty is included (Almeida et al., 2026; De Wolf et al., 2020; Morganti et al., 2025).

These expressions illustrate interface requirements; the review has not fitted or executed a combined material-flow model. Their role is to make the architectural requirements precise enough for subsequent empirical testing.

5.3. How the twin could detect, track and explain uncertainty

The proposed architecture uses a common evidence contract across the four boundaries and allocates different checks to the functions that can perform them. The observation service checks completeness, calibration, coverage and timing. The inference service estimates states or parameters and records posterior or other supported uncertainty. The propagation service passes compatible samples, distributions or scenario identifiers through a stated transformation. The diagnostic service compares predictions with independent observations. The decision service expresses the consequences of the remaining uncertainty for an authorised choice. Identity, provenance and versions connect these services, while stakeholder roles determine who may revise assumptions, approve evidence or act on outputs (Human et al., 2023; Agrawal et al., 2023; Kosse et al., 2025).

A useful diagnostic is a residual

$$r = y_observed - y_predicted$$

. A persistent or unexpectedly large residual can indicate a problem, but may originate in measurement noise, stale input, parameter error, model discrepancy or changed conditions. Diagnosis therefore requires assumptions about both observation and prediction. Under an appropriate predictive model, a standardised residual may divide r by the square root of predictive plus observational variance, including dependence terms where relevant. Such a diagnostic must not be presented as a universal detector or given an arbitrary threshold.

Tracking requires more than storing successive uncertainty widths. A change from a stock interval in tonnes to an impact interval in kilograms CO2-equivalent is a transformation in quantity and unit. Directly comparing the numerical variances would be meaningless. Comparisons of amplification or reduction require a common

quantity, compatible scale and context, or a stated normalisation. Likewise, aggregation can reduce relative variability when errors offset while preserving a shared bias. These distinctions explain why the outcome sketches are a conceptual vocabulary, not numerical results for each stage.

The uncertainty representation should remain plural. Sampling or Bayesian inference is useful when probabilistic assumptions are supported; scenario comparisons remain appropriate when future alternatives do not have defensible probabilities. A qualitative limitation should be carried as a limitation rather than assigned an invented distribution. This allows a stakeholder to see both the calculated uncertainty and what the calculation has left outside its scope (Michiels and Geeraerd, 2020; Kapteyn et al., 2021).

5.4. Why computation, integration and agency have different roles

The architecture in Figures 9–12 makes a specific distinction: uncertainty estimation is a model operation, whereas continuity is also an integration and governance requirement. A sophisticated predictor may quantify its own error while exporting only a point estimate. A semantic platform may preserve provenance while carrying no uncertainty distribution. An automated agent may rerun both without checking whether their units, dependencies and evidence remain compatible. Consequently, no configuration family alone establishes an uncertainty-aware circular workflow.

For GeoAI-to-inventory transfer, computational choices determine the representation of attribute error; internal architecture preserves its association with the building and material estimate. For inventory-to-release transfer, model composition and timing determine whether shared coefficients, lifetime assumptions and renovation events remain connected. For release-to-recovery transfer, inspection and quality evidence must update the recoverable quantity rather than merely annotate a visualization. For recovery-to-impact transfer, the contract must expose substitution, processing and system-boundary assumptions. These allocations are hypotheses for implementation and evaluation derived from the review, not empirical demonstrations of an end-to-end twin.

Agency becomes useful when it makes this process operational: scheduling a check, requesting missing evidence, invoking the appropriate estimator and escalating a decision outside the assessed scope. It cannot replace uncertainty modelling or independent validation. Spatial scalability likewise requires explicit aggregation, dependency treatment and resolution checks; a federated or multiscale label does not demonstrate validity at every scale. The design brief therefore sets the evidence requirements, while functional allocation identifies who or what must satisfy them.

5.5. Limits and research priorities

The strongest quantitative statements in this draft are reproducible frequencies within named coding sets. The archival 17-paper analysis and current 20-paper context analysis use different coding schemes and must not be combined as repeated samples. The former includes secondary and supporting-information-based evidence; the current eligibility ledger contains later targeted full-text checks, but does not retroactively establish every legacy outcome label. The general-DT subset is small and cross-domain. None of these distributions supports a population prevalence estimate or a ranking of uncertainty methods by effectiveness.

No pooled percentage reduction in uncertainty is reported. Such a synthesis would require extracting the quantity of interest, units, probabilistic assumptions, sample or scenario basis, uncertainty metric, upstream/downstream comparison and validation conditions from each eligible primary study. RMSE, coefficient of variation, confidence intervals, prediction intervals and scenario ranges cannot be pooled merely because all describe some form of spread or error. The existing extraction does not provide a harmonised basis for that operation.

The next empirical test should focus on one fully documented boundary before attempting the entire chain. It should compare deterministic transfer with a dependency-preserving alternative, retain identical input evidence, evaluate held-out observations where available, and test how changes affect the decision rather than

interval width alone. Particular priority belongs to S3→S4 and to the connection between recovery quality and environmental substitution. Completing pending eligibility and page-specific evidence reconciliation remains necessary before this working manuscript can be submitted as a closed systematic review.

Taken together, the analyses suggest a precise role for digital twins: maintaining an inspectable relationship between uncertain observations, model transformations and decision consequences over time. The review supports this as an architectural direction, while leaving its end-to-end performance to future validation.

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